INCH-POUND

MIL-DTL-3649H
11 December 2024
SUPERSEDING
MIL-S-3649G(MC)
12 October 1989

DETAIL SPECIFICATION

SHIRT, MAN'S: KHAKI; WITH LONG SLEEVES

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers a khaki, long sleeve, man's shirt made of polyester/wool tropical material (see 6.1).

1.2 Classification. The shirts are of the following classes and sizes, as specified (see 6.2).

1.2.1 Classes.

a. Class 1 – Shirt, polyester/wool, plain cuffs (standard)
b. Class 2 – Shirt, polyester/wool, plain cuffs (optional)
c. Class 3 – Shirt, polyester/wool, French cuffs (optional)

1.2.2 Schedule of sizes. Shirts are of the following neck sizes and sleeve lengths:

Neck size (inches)	Sleeve length (inches)
13	29 through 34 inclusive
13-1/2	29 through 37 inclusive
14	29 through 38 inclusive
14-1/2	29 through 38 inclusive
15	29 through 38 inclusive
15-1/2	29 through 38 inclusive
16	29 through 38 inclusive
16-1/2	30 through 38 inclusive
17	32 through 37 inclusive
17-1/2	32 through 36 inclusive
18	33 through 36 inclusive
18-1/2	33 through 38 inclusive

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-50199 - Thread, Polyester Core, Cotton or Polyester-Covered

A-A-59992 - Cloth, Interfacings: Woven, Cotton and Synthetic

A-A-59994 - Button, Sewing Hole, and Button, Staple, (Plastic)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-21115 - Cloth, Tropical: Wool, Polyester/Wool

MIL-C-29404 - Cloth, Tropical: Polyester and Wool

MIL-DTL-32075 - Label: For Clothing, Equipage, and Tentage, (General Use)

MIL-C-44296 - Cloth, Fusibles

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1492 - Provisions for Evaluating Quality of Men's Shirts

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC TM8 - Colorfastness to Crocking: Crockmeter Method

AATCC TM15 - Colorfastness to Perspiration

AATCC TM20A - Fiber Analysis: Quantitative

AATCC TM117 - Colorfastness to Heat: Dry (Excluding Pressing)

AATCC TM132 - Colorfastness to Drycleaning

AATCC TM158 - Dimensional Changes on Drycleaning, Perchloroethylene

(Copies of these documents are available online at https://www.aatcc.org.)

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ASTM INTERNATIONAL

ASTM D3775	-	Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics
ASTM D3776/D3776M	-	Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D5034	-	Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)
ASTM D6193	-	Standard Practice for Stitches and Seams

(Copies of these documents are available online at https://www.astm.org.)

SAE INTERNATIONAL

AMS-STD-595	-	Colors Used in Government Procurement

(Copies of this document are available online at https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.1.2 Conformance inspection. When specified (see 6.2), a sample shall be subjected to conformance inspection in accordance with 4.4.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Materials and components.

3.3.1 Basic material. The basic material for each class of shirt shall be as specified in 3.3.1.1 and 3.3.1.2. The shade of parts cut from ends (see Table V, operation 1.b) shall be a good match to the shade of the main assembly.

3.3.1.1 Class 1 shirt. The basic material for the Class 1 shirt shall be polyester/wool tropical cloth conforming to MIL-C-29404, Type I, when tested as specified in 4.5.1. The shade shall be USMC Khaki M-1.

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3.3.1.2 Class 2 shirt. The basic material for the Class 2 shirt shall be 55 percent polyester/45 percent wool tropical cloth conforming to MIL-DTL-21115, Type III, Class 3, when tested as specified in 4.5.1. The shade shall be USMC Khaki M-1.

3.3.2 Interlining (interfacing) facing. The material for interfacing the collar, collar stand, Class 1 and 2 cuffs, pocket flaps, and front shall be polyester cloth conforming to A-A-59992, Type II, Class 4 when tested as specified in 4.5.1. As an alternate, a non-woven fusible conforming to MIL-C-44296, Type V, Class 2 or 3, Style B may be used. The color of the interfacing shall be natural.

NOTE: The terms interlining and interfacing are used interchangeably in this specification.

3.3.3 Collar stand lining for class 2 and 3 shirts. The material for the inner ply of the collar stands of the Class 2 and 3 shirts shall be an acetate lining material, twill or satin (alternate) weave, conforming to the requirements specified in Table I when tested as specified in 4.4.1.

TABLE I. Lining requirements.

Characteristic	Twill requirement	Satin requirement
Shade	Good match to the shade of the basic material	
Fiber identification	Acetate	
Yarns per inch (minimum)		
Warp	120	288
Filling	68	68
Weight (minimum)	3.8 oz/yd^2	4.9 oz/yd^2
Breaking strength, pounds (minimum)		
Dry: warp	74	83
Dry: filling	49	117
Wet: warp	54	54
Wet: filling	34	77
Colorfastness to perspiration, rating (minimum) (see 6.4)		
Color change	5	5
Staining on cotton	3.5	5
Staining on wool	4.5	5

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TABLE I. Lining requirements – Continued.

Characteristic	Twill requirement	Satin requirement
Colorfastness to drycleaning, rating (minimum) (see 6.4)		
Color change	4.5	5
Staining on cotton	4.5	5
Staining on wool	4.5	5
Colorfastness to crocking, rating (minimum) (see 6.4)	4.5	4.5
Dimensional stability to drycleaning, percent (maximum)		
Individual sample unit	±3.5%	±3.5%
Lot average	±3.0%	±3.0%

3.3.4 Thread. The threads used in the construction of the shirt shall conform to the requirements of A-A-50199, type I when tested as specified in 4.5.1. Threads shall match the shade requirement specified in Table II.

TABLE II. Thread requirements.

Use (seam type)	Tex	Ply	Shade
General seaming and overedging	31-35 for needle and bobbin	2	Good match to the shade of the basic material (see 3.3.1 and 6.5)
Button and buttonhole	46-60 for needle and bobbin	2 or 3	

3.3.5 Buttons. The buttons shall conform to A-A-59994, Type I, Class 1, Style 24, 20 ligne when tested as specified in 4.5.1. The color of the buttons (see 6.3.1) shall match Khaki 475, color chip 23420 of AMS-STD-595, with a glossy finish.

3.3.6 Collar stays. The collar stays shall be made from an industry standard material of 0.010 ± 0.001 inch when tested as specified in 4.5.1 that will withstand dry cleaning. The color shall be natural (clear or opaque). The stay shall measure 1/4 to 3/8 inches wide and 2-1/4 to 2-5/16 inches long. Both ends shall be rounded, or one end shall be rounded and the other end cut at an angle with the point rounded.

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3.3.7 Labels.

3.3.7.1 Combination identification and instruction label. Each shirt shall have a combination identification and instruction label conforming to MIL-DTL-32075, Type VI, Class 15 when tested as specified in 4.5.1. The label shall measure not less than 2-1/2 inches wide by 1-1/2 inches long. The label shall be printed legibly with the following information:

SHIRT, MAN'S, KHAKI LONG SLEEVES
CONTRACT NO. SPE1C1-00-0-0000 (EXAMPLE)
NAME OF CONTRACTOR:
FIBER CONTENT: 55% POLYESTER/45% WOOL (EXAMPLE)
DRYCLEAN OR LAUNDER BY HAND.
CAUTION: BLEACH, FABRIC SOFTENER, DRYER SHEETS, STARCH AND/OR HOT WATER (EXCEEDING 100 °F) WILL HARM THE FABRIC OF THE SHIRT.

3.3.7.2 Alternate combination identification and instruction label and alternate size label. As an alternate to the printed labels, the identification and instruction information and the size information may be marked on the shirt by heat transfer, direct printing, or stamping. The marking shall conform to MIL-DTL-32075, Type III or IV, Class 15 when tested as specified in 4.5.1. The printing shall have a rating of 3.5 minimum for colorfastness to laundering and drycleaning when tested in accordance with MIL-DTL-32075.

3.3.7.3 Size label, cloth, printed. The size label shall conform to MIL-DTL-32075, Type VI, Class 2 and shall measure not less than 1-1/4 inches wide by 1-1/2 inches long. The size designation, consisting of the neck size, sleeve length, and the National Stock Number (NSN), shall be indicated on the label as shown in the following example. The NSN may be omitted from the size label, but when omitted, the NSN shall be included in the combination identification and instruction label.

15-33

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3.3.7.4 Barcode label. Each shirt shall be individually barcoded with a paper label/tag conforming to MIL-DTL-32075, Type VIII, Class 17 when tested as specified in 4.5.1. The tag shall be attached to the item by a fastener. The label's location and attachment method shall cause no damage to the item. The barcoding element shall be a 13-digit NSN and the appropriate 12-digit Universal Product Code (UPC) assigned to the NSN by the Government. The initials "UPC" shall appear beneath the code. The barcodes for the NSN and UPC shall be a medium to high density and shall be located so that they are completely visible on the item when the item is folded or packaged as specified herein.

3.4 Design. The shirts are in the United States Marine Corps design (see Figure 1 and Figure 2). The shirt is semi-form fitting at the waist and has a six-button front placket, two patch-type chest pockets with buttoned flaps, a two-piece back yoke, a collar and collar stand, and long sleeves with cuffs and a buttoned sleeve placket. The Class 1 and 2 shirts have plain cuffs, and the Class 3 shirts have French cuffs. The collar stand for Class 1 shirts is lined with self-fabric, and the collar stand for Class 2 and 3 shirts is lined with an acetate material. Both classes of shirts have sharp, permanent creases on the front and back of the shirts (see 3.4.1).

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3.4.1 Permanent creases. Each shirt shall have three sharp, vertical, permanent creases on the shirt back (excluding the yoke) and two sharp, vertical, permanent creases on the shirt front including sharp, permanent creases through the front pockets and front pocket flaps. The sleeves shall also have a permanent crease along the top, opposite the seam. Each permanent crease shall be a single, continuous line formed by applying a 1/16-inch (nominal) non-staining, silicone adhesive bead to the back side of the basic material. The silicone bead shall be applied to the following cut parts prior to assembly: back, fronts, pockets, sleeves, and top pocket flaps. Each bead shall be applied evenly and equally in a straight, vertical line from the top to the bottom of each cut part. The finished permanent creases shall show no evidence of flaking, decomposition, failure to retain creases, cracking, staining, strike back onto face of material, or increased stiffness. The creases shall have a minimum rating of "3" when evaluated for permanent crease resistance to dry heat (3 exposures) and steam pressing (3 exposures) and perchloroethylene solvent when tested as specified in 4.5.5.1, and shall have a minimum rating of "pass" for resistance to hot water when tested as specified in 4.5.5.2.

3.5 Figures. Figure 1 and Figure 2 are provided for information only. When inconsistencies exist between the written specification and the figures, the written specification shall govern.

3.6 Patterns. Standard patterns will be furnished by the Government (see 6.2). Government patterns show size, seam allowances, and directional lines for cutting, and are marked or notched for proper placement and assembly of component parts. The directional lines indicate the warp direction of the cloth unless otherwise indicated on the patterns. The list of pattern parts and computer nomenclature in 3.6.1 is provided to ensure that the pattern set provided is complete. The Government patterns shall not be altered in any way and shall only be used for cutting the contractor's working patterns. The working patterns shall be identical to the Government patterns, except that additional notching, if needed to facilitate manufacture, is permitted on the contractor's working patterns. Minor modifications are permitted where necessary to accommodate the manufacturer's processes and the use of automated equipment, provided the modifications do not alter the dimensional, serviceability, or appearance requirements cited in this document or governed by the Government pattern.

3.6.1 Pattern parts. Component parts shall be cut from the materials specified and in accordance with the number of parts specified in Table III. Requirements for cutting component parts that are not included in Table III are included in 3.8.

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TABLE III. Pattern parts.

Material	Nomenclature	Cut parts
Basic material (3.3.1)	Back	1
	Front	2
	Yoke	2
	Sleeve	2
	Cuff class 1&2	4
	French cuff class 3	4
	Collar	2
	Collar stand	2 1/
	Stay holder	2
	Pocket	2
	Flap	4 2/
	Sleeve placket	2
Interfacing (3.3.2)	Collar interfacing	1
	Collar stand interfacing	1
	Pocket flap interfacing	2 2/
	Cuff interfacing class 1&2	2
	Front interfacing	1
Shaper	Finish pocket	-
	Finish flap	-
	Sleeve placket template	-

1/ For Class 2 and 3 shirts, the same pattern part is used to cut the outer ply of the collar stand from basic material (cut 1) and to cut the inner ply of the collar stand from lining material (cut 1).

2/ The same pattern part is used to cut the pocket flap interfacing (cut 2).

3.7 Construction. Construction of the shirt shall conform to the requirements specified in Table IVand be manufactured in accordance with 3.8. Stitch and seam types shall be as specified in ASTM D6193.

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TABLE IV. Construction requirements.

Construction feature	Requirement
1. Thread	
a. Thread type and size	Table II
b. Thread tension	Maintained so that there is no tight, loose, or unbalanced stitching.
c. Thread shade on outside of shirt	Consistent thread shade on the outside of the shirt. Multiple shades of thread on the outside of the shirt are not allowed.
d. Thread breaks (all stitch types)	Secured by stitching back of the break not less than 1/2 inch.
	If 401 stitch type, repaired using 301 stitch type.
2. Stitches and seams	
a. Stitch and seam types	See 3.8.
b. Stitches per inch	See 3.8.
c. 401 stitch type	Loops of the chain finish on the inside of the shirt.
d. Overedging	Knife set to trim the loose threads and raveled edge without cutting the material.
e. Seams and seam allowances	Maintained with seam uniformly sewn so that no raw edges, runoffs, twists, pleats, puckers, gathers, other distortions, or open seams result.
f. Beginning and ending of seams	Start and finish evenly (see 2.h and 2.i).
g. Joined components	Lie smooth and flat.
h. Ends of seams and stitching – 301 stitch type	If not caught in another line of stitching: backstitched not less than 1/4 inch.
	If the end of a continuous line of stitching (except label): overlapped not less than 1/2 inch.
	If label stitching: overlapped not less than 3 stitches.
i. Ends of seams and stitching – 401 stitch type	Caught in other seams or stitching.
j. Skipped stitches (301 or 401)	Repaired by using 301 stitch type.
k. Edge, raise, or topstitching	Positioned 1/16 inch from the edge.
l. Additional bastings	Acceptable provided the basting thread does not show on the outside of the finished item.

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3.8 Manufacturing operations requirements. Following the exact sequence of operations specified in Table V is not required provided the finished shirts meet the finished form, fit, and function requirements as those produced by following the sequence of operations as specified in Table V. Stitch and seam types shall be as specified in ASTM D6193. Minor modifications are permitted where necessary when using automated equipment provided the finished shirts meet the finished form, fit, and function requirements as those produced by following the operations as specified in Table V. Pressing shall be accomplished with a heated pressing iron or pressing machine to obtain the desired appearance in accordance with manufacturer suggested procedures. A steam-inflated bag-type machine shall not be used. Any additional underpressing to facilitate manufacturing is permissible.

TABLE V. Table of operations.

No.	Operation	Stitch type	Seam type	Stitches per inch
1.	Cut and shade mark.			
	a. Cut the parts (see 3.6.1) as indicated on the patterns provided and in the material specified. Additional cutting requirements are provided below for those components not controlled by a furnished pattern piece.			
	b. Cut all basic material (see 3.3.1) parts from one piece of material except for the following, which may be cut from ends: under ply of collar, under ply of plain cuffs, and under ply of pocket flaps. The shade of all parts cut from ends is a good match to the shade of the main assembly.			
	c. Cut the lining material (see 3.3.2) for the class 2 and 3 collar stands (inner ply) in accordance with the pattern.			
	d. Cut the interfacing parts (see 3.3.3). As an alternate, a fusible interfacing (see 3.3.3) is allowed for the interfacing parts. The interfacing for the collar and collar stand is cut on the straight (warp or filling) direction in accordance with the patterns. The fusible is applied in accordance with the manufacturer's recommended practice.			
	e. The use of drill holes is prohibited.			

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TABLE V. Table of operations – Continued.

No.	Operation	Stitch type	Seam type	Stitches per inch
1.	f. Mark or ticket the component parts of the shirt to ensure a uniform shade and size throughout the garment. Ensure the numbering does not show on the outside of the finished garment and, wherever possible, is covered by the seam allowance. The use of an ink pad numbering machine, rubber stamp, or pencil is allowed. The following methods are not allowed: corrosive metal fastening devices, sew-on shade tickets, any means of attachment that leaves a hole after removal, and adhesive-type tickets which discolor or leave adhesive on the material upon removal of tickets.			
2.	Replace defective or damaged parts.			
	a. During cutting, replace all parts containing defects such as a hole, cut, tear, smash, mend, burn, scorched area, open place, loose yarn, snagged thread, area of no dye penetration, dye streak, missing yarn, broken yarn, thin place, shade bar, or woven in waste.			
	b. During the manufacturing process, replace all parts with damage such as, but not limited to: needle chews, scissor or knife cuts, tears, holes, mends, or burns.			
3.	Make pockets and permanent creases on pockets.			
	a. Finished appearance: The hem of the pocket is uniformly stitched and finishes smooth and flat. The chest pockets are uniform in size and shape in accordance with the patterns. The permanent crease is a sharp, continuous, even line centered in the pocket width and is uniform in appearance.			
	b. Hard press and crease the pocket from the top of the pocket to the bottom of the pocket in a straight, vertical line centered in the pocket width as indicated by the marks on the patterns.			
	c. Apply an even, continuous, 1/16-inch wide silicone bead to the crease in the pocket on the back side of the basic material.			

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TABLE V. Table of operations – Continued.

No.	Operation	Stitch type	Seam type	Stitches per inch
3.	d. Fold the pocket hem in accordance with the fold line indicated on the pattern and hem with two rows of stitching, 1/4 inch gage, with the outer row 1/16 inch from the folded edge.	301 or 401	EFa-2	12-16
	e. Turn under and crease the side and bottom edges of the pocket in an edge creasing machine in accordance with the finished shaper. The creased pocket shall conform to the finished pocket shaper.			
4.	Make pocket flaps and permanent creases on pocket flaps.			
	a. Finished appearance: The pocket flap is uniformly stitched with the seams worked out to the edges. The flap finishes smooth and flat without twists, pleats, puckers, or raw edges. The flaps are uniform in shape and size in accordance with the patterns. The buttonholes are securely tacked at each end. The sharp, permanent crease is a continuous, even line, centered in the pocket flap width and is uniform in appearance.			
	b. The flap is made of two plies of basic material and one ply of interfacing. If used, the alternate fusible interfacing is applied to the underside of the outer ply.			
	c. Hard press and crease the outer ply of the pocket flap from the top to the bottom of the pocket flap in a straight, vertical line, centered in the pocket flap width, as indicated by the marks on the pattern.			
	d. Apply an even, continuous, 1/16-inch wide silicone bead to the crease on the back side of the outer ply pocket flap.			
	e. Seam all plies together 1/4 inch from the side and bottom edges. Trim the corners and points for turning.	301	SSe-2(a)	12-16
	f. Turn the flap, working the corners and points out completely, and working the seams out to the edges in accordance with the finished pocket flap shaper. Edgestitch the flap uniformly 1/8 inch from the seamed edges. The seam finishes at the edges of the flap.	301	SSe-2(b)	12-16
	g. Trim the top edge of the flap evenly.			

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TABLE V. Table of operations – Continued.

No.	Operation	Stitch type	Seam type	Stitches per inch
4.	h. Make a vertical, straight-type buttonhole, cut 5/8 inch long, through the center of each flap, securely tacking each end. The lower end of the buttonhole finishes 5/8 to 1/2 inch above the center point. The buttonhole purling finishes on the outside of the flap.	Buttonhole		55 per buttonhole including tack
5.	Make cuffs.			
	a. Finished appearance: The cuffs are uniformly stitched and finish smooth and flat without twists, puckers, or pleats. The cuffs are uniform in shape with rounded corners.			
	b. For class 1 and 2, fold the top edge of the outer ply of the cuff over the top edge of the interfacing 3/8 inch and single stitch 1/4 inch from folded edge, catching the interfacing in the stitching.	301	EFa-1	12-16
	c. Seam the respective plies together along the bottom and side edges with a 1/4-inch seam.	301	SSe-2(a)	12-16
	d. Turn and edgestitch 1/2 inch from folded edge. The edge stitching finishes 1/2 inch from the top edge of the finished cuff. The edgestitching may taper to the edge of the cuffs at the ends of the stitching and is not required to be backstitched.	301	SSe-2(b)	12-16
	e. As an alternate, the class 1 and 2 cuffs may be fused.			
6.	Make collar and collar stand.			
	a. Finished appearance: The collar is uniformly stitched and finishes smooth and flat without twists, puckers, pleats, or other distortions. The points are worked out completely and are uniform in length. The seams are worked to the edge of the collar. Note: Collar stays must be placed in the position designated on the pattern part to accommodate the required insignia.			
	b. The collar consists of two plies of basic material, a single ply of interfacing, and two collar stay holder pieces with the collar stays attached. As an alternate, the topcollar may be fused.			

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TABLE V. Table of operations – Continued.

No.	Operation	Stitch type	Seam type	Stitches per inch
6.	c. Sew or spot laminate the collar stays to the stay holder pieces in accordance with the marks on pattern. The spot lamination extends not less than two-thirds the stay length.	301	SSa-1	10-14
	d. As an alternate, the collar stays may be spot laminated to the interfacing, which eliminates the use of collar stay holder pieces.			
	e. Properly position the undercollar, interfacing, stay holder piece assembly (when used) and topcollar together and seam with a 1/4-inch seam from the top and side edges. The stay holder piece assembly shall be positioned between the undercollar and the interfacing on the finished collar.	301	SSe-2(a)	12-16
	f. Trim the points and turn collar on a heated collar point turning and pressing machine, working the points out completely and working the seams to the edges.			
	g. Edgestitch the collar uniformly 1/8 inch from the seamed edge.	301	SSe-2(b)	12-16
	h. Trim the collar even at the lower edge.			
7.	Make collar stand and attach to collar.			
	a. Finished appearance: The collar stand is joined to the collar without pleats, twists, puckers, or other distortions. The ends of the collar stand are rounded and well formed.			
	b. Position the interfacing and outer ply of collar stand with notches matching. Stitch the two plies together from end to end with four horizontal rows of stitching, evenly distributing the fullness of the interfacing between the rows of stitching. The top and bottom row shall be 1/2 to 5/8 inch from the top and bottom raw edges of the collar stand respectively, and the remaining two rows shall be evenly spaced between the top and bottom rows. As an alternate, the outer ply of the collar stand may be fused.	301	OSa-4	12-16

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TABLE V. Table of operations – Continued.

No.	Operation	Stitch type	Seam type	Stitches per inch
7.	c. Position the collar stand interfacing assembly on the outside of the undercollar, and the inner ply of the collar stand (basic material for class 1 and lining material for classes 2 and 3) on the outside of the topcollar, with the notches matching, and seam all plies together 1/4 inch from the lower edges; continue the stitching to the ends of the collar stand, seaming together both plies of the collar stand.	301	SSa-1	12-16
	d. Turn down the collar stand pieces and stitch 1/16 inch from the raised edge.	301	SSq-2 (raised edge shows finished seam)	12-16
8.	Bind sleeve opening.			
	a. The binding is seamed to the back edge of the sleeve opening without twists, pleats, puckers, or raw edges, and finishes smooth, flat, and uniform in appearance. The binding finishes 1±1/16 inch wide.			
	b. Bind the raw edge of the sleeve at the back opening with a 1-1/2-inch wide strip of basic material, with the raw edges of the binding turned under and caught in the stitching 1/16 inch from the inner folded edge.	301	BSc-1	12-16
9.	Seam placket to sleeve.			
	a. Finished appearance: The placket is seamed to the front edge of the sleeve opening without twists, pleats, or puckers, and finishes smooth, flat, and uniform in appearance. The placket finishes 1±1/8 inch wide on the outside.			
	b. Turn under and crease the side edges and upper end of placket pieces on a placket folding and pressing machine, forming a point at the upper end.			
	c. Position the placket piece on the inside of the sleeve at the front edge of the opening with the raw edges even and seam 1/4 inch from the raw edge. The stitching shall extend from the bottom to top end of the opening.	301	LSq-2(a)	12-16
	d. Turn the placket at the seam and topstitch the placket 1/16 inch from the raised edge.	301	LSq-2(b)	12-16

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TABLE V. Table of operations – Continued.

No.	Operation	Stitch type	Seam type	Stitches per inch
9.	e. Tack the placket, the binding, and the tongue notch together at the top end of the sleeve opening.	301	SSa-1	12-16
	f. Turn the placket to the outside of the back folded edge and stitch the placket to the body of the sleeve 1/16 inch from the folded front and slanted top edges and continue stitching along the back edge from a distance of 1-1/2 ± 1/8 inches from the point, lapping the placket on the bound edge of the sleeve opening.	301	LSd-1	12-16
	g. Stitch and backstitch across the width of the placket from the end of the stitching at the back edge to the row of stitching at the front edge. The stitching shall be through the placket and the bound edge of the sleeve opening.	301	SSa-1	12-16
10.	Join yoke to back and make permanent creases in back.			
	a. Finished appearance: The yoke is seamed to the back without pleats, puckers, twists, raw edges, or other distortions, and is uniform in appearance and placement. Each permanent crease is a continuous even line of uniform appearance.			
	b. Hard press and crease the back from the top to the bottom of back with three straight vertical creases positioned in accordance with the marks on the pattern (one crease at the center of the back and one on each side of the center equidistant from the center crease).			
	c. Apply an even, continuous 1/16-inch bead of silicone to the center of each of the three pre-pressed creases on the back side of the basic material.			
	d. Position the top edge of the back between the two lower plies of the yoke, with the edge even and stitch together with a 1/2-inch seam. Turn the yoke to the finished position and topstitch the yoke with 1/4 inch gage. The outer row of the stitching shall be 1/16 inch from the folded edge.	301	SSq-3(a) and SS-3(b)	12-16
11.	Make permanent creases in fronts.			
	a. Hard press and crease each front from the top to the bottom with a straight vertical crease in accordance with the marks on the pattern.			

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TABLE V. Table of operations – Continued.

No.	Operation	Stitch type	Seam type	Stitches per inch
11.	b. Apply an even, continuous, 1/16-inch silicone bead to the center of the pre-pressed crease in each front on the back side of the basic material.			
12.	Make button stand and buttonhole placket.			
	a. Finished appearance: The button stand and buttonhole placket are uniformly stitched and finish smooth and flat without pleats, puckers, twists, or other distortions.			
	b. Form the button stand by folding the right front edge 2±1/8 inches to the outside, turning under the raw edge 3/4 inch and stitching 1/16 inch from the inner folded edge. The button stand shall finish 1-1/4±1/8 inches wide.	301 or 401	EFb-1	12-16
	c. Attach the front interfacing on the outside left front. Form the buttonhole placket by folding the left front edge to the outside 1-7/8±1/8 inches, turning under the raw edge 5/16 to 3/8 inch and stitching 3/16 inch from each folded edge. The buttonhole placket shall finish 1-1/2±1/8 inches wide.	301 or 401	EFk-2	12-16
13.	Attach label.			
	a. The combination identification and instruction label is attached on the outside of the shirt approximately 2-1/2 inches from the front folded edge and no more than 1-1/2 inches up from the bottom finished edge of the shirt on the left side as worn. Stitch the label on all four sides 1/16 to 3/32 inch from the edges. The stitching shall not be through the printing. As an alternate, the label may be marked directly on the shirts (see 3.3.7.2 and operation 23).	301	LSbj-1	8-12

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TABLE V. Table of operations – Continued.

No.	Operation	Stitch type	Seam type	Stitches per inch
14.	Set pockets and pocket flaps.			
	a. Finished appearance: The pockets and pocket flaps are uniform in appearance and placement and lie smooth and flat without pleats, puckers, twists, or raw edges. The pockets and pocket flaps are parallel to the front edge of shirt when buttoned. The pocket and flap, on left and right sides respectively, shall be in alignment with each other. The permanent crease on the pocket flap, pocket and front are vertically aligned, and exhibit a single, continuous, unbroken line on each front. The pockets and flaps conform to the measurement shown in Table VII.			
	b. Mark the fronts for position of the pockets and pocket flaps, as indicated by the patterns.			
	c. Position the raw edge of the pocket flap in line with the mark on the front. Seam the pocket flap to the front 1/8 inch from the raw edge.	301	LSbl-2(a)	12-16
	d. Turn down the flap at the seam and topstitch 1/4 inch from the raised edge.	301	LSbl-2(b)	12-16
	e. Position the pre-creased pocket at the mark on the front. Seam the pocket to the front uniformly 1/16 inch from the side and bottom turned edges, securely tacking the top corners.	301	LSd-1	12-16
15.	Seam shoulders.			
	a. Finished appearance: The shoulder seam is uniformly stitched and finishes evenly without pleats, puckers, twists, or raw edges. The back is lapped on the fronts.			
	b. With a 1/2-inch seam allowance, join the yoke and front together at the shoulders with a double-lapped and double-stitched seam, 1/4 inch gage, with the outer row 1/16 inch from the folded edge.	301 or 401	LSc-2	12-16

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TABLE V. Table of operations – Continued.

No.	Operation	Stitch type	Seam type	Stitches per inch
16.	Set collar and attach size label.			
	a. Finished appearance: The collar is uniformly stitched to the shirt without pleats, puckers, twists, or raw edges. On the finished shirt, the ends of the collar shall be 3-3/8 +3/16, –1/8 inches long when the collar is turned up and measured from the juncture point of the collar and the collar stand to the collar point. The seam joining the collar to the body of the shirt is not visible below the finished edge of the collar when the collar is properly folded and pressed at the breakline. The cloth size label, when used, is properly positioned and caught in the seam joining the collar stand to the shirt.			
	b. Seam the collar stand to the neck of the shirt on the outside, catching the interfacing in the stitching.	301	SSa-1	12-16
	c. Turn in the bottom edge of the collar stand and single stitch through the stand and shirt, catching the top edge of the label. Continue the stitching around the ends of the stand with the stitching superimposed on the raised stitching of the collar stand (operation 7.d).	301	LSd-1 and SSe-2(b)	
17.	Set sleeves.			
	a. Finished appearance: The sleeve is set properly and uniformly stitched without pleats, puckers, twists, or raw edges.			
	b. Trim the yoke even with the back part at the armhole, if necessary.			
	c. With a 3/8-inch seam allowance, join the sleeves to the shirt body by overedging and seaming the raw edges together in one operation.	515, 516, 517, 518, or 519	SSa-2	10-14
	d. Topstitch around the armhole with two rows of topstitching 1/16 inch from the edge, 1/4 inch gage.	301 or 401	LSq-2	12-16

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TABLE V. Table of operations – Continued.

No.	Operation	Stitch type	Seam type	Stitches per inch
18.	Close sides and sleeves.			
	a. Finished appearance: The sleeve and side seams are uniformly stitched without pleats, puckers, twists or raw edges. The seams start and finish evenly. The ends of the armhole seams are not staggered more than 3/8 inch. The back is lapped on the fronts, or the fronts on the back, but not both within the same shirt.			
	b. With a 1/2-inch seam allowance, join the fronts to the back part at the sides and close the sleeve underarm with a double lapped, double-stitched seam, staggering the ends of the armhole seam. The double row of stitching is 1/4 inch gage with the outer row 1/16 inch from the folded edge.	301 or 401	LSc-2	12-16
19.	Hem bottom of shirts.			
	a. Finished appearance: The bottom hems are uniformly stitched without twists, puckers, pleats or other distortions, and finish 3/16 to 1/4 inch wide.			
	b. Turn the bottom of the fronts and back of the shirt with the raw edge turned in and single stitched through the shirt, 1/16 inch from the upper folded edge, forming the hem.	301	EFb-1	12-16
20.	Join cuffs to sleeves.			
	a. Finished appearance: On the finished shirts, the plain cuff (classes 1 and 2) is 2-1/2 to 2-5/8 inches wide and the French cuff (class 3) is 4-7/8 to 5-1/8 inches wide.			
	b. Classes 1 and 2: Position the cuff on the lower inside end of sleeve with the raw edges even. The fullness of the sleeve bottom shall be taken up with one pleat facing the buttonhole side of the sleeve placket. The pleat is positioned and formed in accordance with the marks on the patterns. Join the underply of the cuff and the end of the sleeve together with a 1/4-inch seam.	301	LSq-2(a)	12-16
	c. Turn the cuff to the outside with the hemmed edge lapped on the outside of the sleeve. Stitch the top ply to the sleeve 1/16 inch from the finished top edge.	301	LSb-1	12-16

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TABLE V. Table of operations – Continued.

No.	Operation	Stitch type	Seam type	Stitches per inch
20.	d. As an alternate, the cuffs may be attached in one operation provided the finished appearance is the same as specified in operations 20.b and 20.c above, and the ends of the rows of stitching are backtacked or crossed with another line of stitching.	301	LSc-1	12-16
	e. Class 3: Position the cuff on the lower inside end of sleeve with the raw edges even. Join the underply of the cuff and the end of the sleeve together with a 3/8-inch seam, turning the bound edge of the sleeve opening flat against the inside of the sleeve and catching the end in the stitching.	301	LSq-2(a)	12-16
	f. Raise the cuff and topstitch the underply 1/16 inch from the raised seam.	301	LSb-1	12-16
	g. Turn the cuff to the outside, lapping the top ply on the outside of the sleeve. Stitch the top ply to the sleeve with the upper edge turned under and caught in the stitching 1/16 inch from the upper folded edge.	301	LSb-1	12-16
	h. Edgestitch the cuffs 3/16 inch from side and bottom seamed edges.	301	SSe-2(b) shows finished seam	12-16
21.	Make buttonholes.			
	a. Finished appearance: The buttonholes are clean-cut, well-made, and correctly positioned. The buttonholes are straight-type, cut 5/8 inch long, and fit a 20-ligne sewing hole button. The purling is on the outside of the shirt. The stitching is securely caught in the fabric.			

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TABLE V. Table of operations – Continued.

No.	Operation	Stitch type	Seam type	Stitches per inch
21.	b. The front buttonholes are midway between the folded edges of the buttonhole placket and positioned according to the marks on the patterns. The front end of the collar stand buttonhole extends 1/8 inch ahead of the line of the center placket buttonholes. The Class 1 and 2 cuff buttonholes are midway between the top and bottom edges of the cuff with the back end of the buttonhole 3/8 to 1/2 inch from the end of the cuff. The class 3 cuff buttonholes are spaced 3/4±1/8 inch apart with the bottom buttonhole 1-3/8 to 1-1/2 inches above the bottom folded edge of the cuff and the outer end of the buttonhole 3/8 to 1/2 inch from the end of the cuff at the opening.			
	c. Make vertical straight-type buttonholes at each mark on the buttonhole placket and sleeve placket and make horizontal straight-type buttonholes on the collar stand and on the cuffs in accordance with the marks on the pattern, securely tacking each end of the buttonhole.	Buttonhole or 101 (zigzag)		55 per buttonhole including tack
22.	Attach buttons.			
	a. The buttons are securely sewn and properly engage the buttonholes in the buttonhole placket, collar stand, sleeve placket, pocket flaps, and class 1 and class 2 cuffs.			
	b. Sew five 20-ligne sewing hole buttons to the button stand through the right front, with the center of the buttons 5/8 to 3/4 inch from the front edge of the button stand.	101 or 301		21-24 per button 16-18 per button
	c. Sew one 20-ligne sewing hole button to the right end of the collar stand, with the center of the button aligned with the center of the front buttons.	101 or 301		21-24 per button 16-18 per button
	d. Sew one 20-ligne sewing hole button to the bound edge of each sleeve placket.	101 or 301		21-24 per button 16-18 per button

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TABLE V. Table of operations – Continued.

No.	Operation	Stitch type	Seam type	Stitches per inch
22.	e. Class 1 and 2: sew one 20-ligne sewing hole button to the back end of each cuff.	101 or 301		21-24 per button 16-18 per button
	f. Sew one 20-ligne sewing hole button to each pocket through the hem.	101 or 301		21-24 per button 16-18 per button
23.	Identification and instruction marking and size marking.			
	a. When marking is used in lieu of a cloth identification and instruction label and a size label, the identification and instruction marking is applied to the shirt at the location specified for the cloth label (see operation 13), and the size is marked on the inner collar stand at the center back.			
24.	Cleaning.			
	a. Trim all ends of stitching and remove the loose ends from the shirt.			
	b. Remove all spots, stains, and shared tickets or markings without injury to the material.			
25.	Pressing.			
	a. Finished appearance: the entire shirt is hard-pressed.			
	b. Press collar on a collar machine, creasing on the breakline and shaping to obtain the proper contour while pressing.			
	c. Press buttonhole placket and button stand smooth and flat.			
	d. Hard press entire shirt (fronts, back, sleeves, pocket flaps and pockets). Button the fronts, pocket flaps, sleeve plackets and cuffs.			

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3.9 Finished dimensions. The shirts shall conform to the measurements in Table VI and Table VII as defined in Table VIII.

TABLE VI. Finished shirt measurements (inches).

Neck size	Collar length (A)	Chest (B)	Back length (C)	Sleeve length (D)
13	13-1/2	35	30-3/4	1/
13-1/2	14	37	30-3/4	1/
14	14-1/2	39	30-3/4	1/
14-1/2	15	41	30-3/4	1/
15	15-1/2	43	31-1/4	1/
15-1/2	16	45	31-1/4	1/
16	16-1/2	47	31-1/4	1/
16-1/2	17	49	31-1/4	1/
17	17-1/2	51	31-1/4	1/
17-1/2	18	53	31-1/4	1/
18	18-1/2	55	31-1/4	1/
18-1/2	19	57	31-1/4	1/
Tolerance	+1/4, –1/8	±1/2	+1, –1/2	+1/4, –1/2

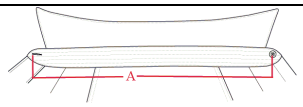
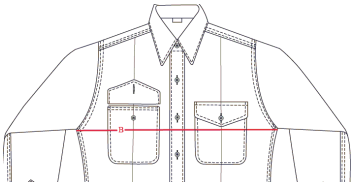
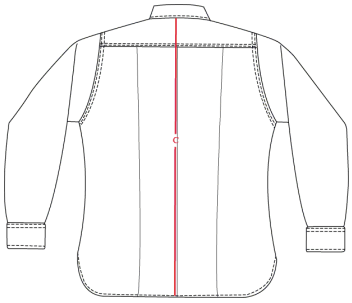
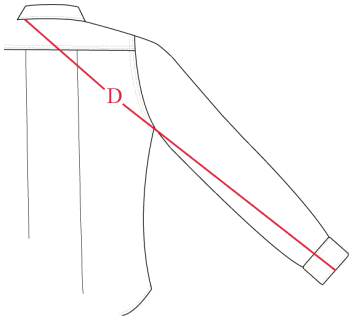
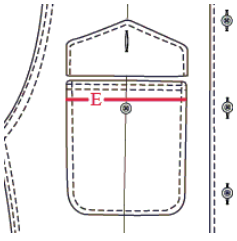
1/ Sleeve length is as specified in section 1.2, as applicable to the size shirt being produced.

TABLE VII. Pocket and flap measurements (inches).

Pocket and flap dimension	Measurement for neck sizes 13 to 14-1/2 (inclusive)	Measurement for neck sizes 15 to 18-1/2 (inclusive)
Pocket width (E)	4-1/2	5-1/8
Pocket length (F)	5-3/4	6
Flap width (G)	4-3/4	5-3/8
Flap length at sides (H)	1-5/8	1-7/8
Flap length at center (I)	2-1/2	2-7/8
Overall length of pocket and flap (J)	6-1/4	6-1/2
Tolerance	±1/8	±1/8

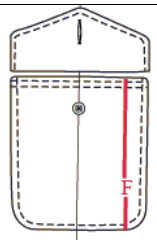
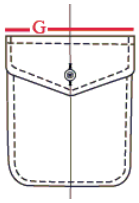
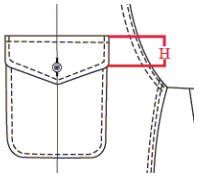
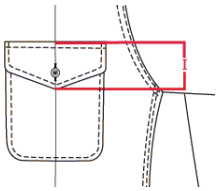
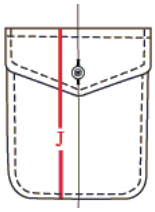
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TABLE VIII. Methods of measure.

Letter	Measurement	Method	Description
A	Collar length	With the collar unbuttoned and laid flat, from the front cut end of the buttonhole to the center of the button	
B	Chest	Twice the measurement from folded edge to folded edge in line with the pit of the armhole with the shirt buttoned and laid flat	
C	Back length	From the lower collar stand joining seam to the bottom edge of the shirt at the center back	
D	Sleeve length	Class 1 and 2: with the sleeve extended, from the center back at the lower edge of the undercollar to the lower edge of the cuff Class 3: cuff is folded with the buttonholes matching. With the sleeve extended, from the center back at the lower edge of the undercollar to the lower folded edge of the cuff	
E	Pocket width	From side edge to side edge across the top of the pocket	

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TABLE VIII. Methods of measure – Continued.

Letter	Measurement	Method	Description
F	Pocket length	From the top edge to the bottom edge of the pocket	
G	Flap width	From side edge to side edge across the top edge of the flap	
H	Flap length at sides	From the top edge to the bottom edge of the flap at the side edge	
I	Flap length at center	From the top edge to the bottom point of the flap at the center	
J	Overall pocket length	From the top edge of the flap to the bottom edge of the pocket at the center of the pocket	

3.10 Workmanship. Each finished shirt shall conform to the quality of product established by this document. Defects shall not exceed the established maximum acceptance quality limits (see 6.2) and shall not adversely affect the serviceability, appearance, and uniformity of the product.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in applicable test method documents.

4.3 First article inspection. The first article, in accordance with 3.1.1, shall be examined in accordance with 4.5.1 through 4.5.5.2 for compliance with design, configuration, workmanship, dimensional requirements, and testing as specified in this document.

4.3.1 First article samples and acceptance criteria. Unless otherwise specified in the contract (see 6.2), first article samples shall be provided. The sample size shall be as specified (see 6.2). The sample unit shall be one finished shirt. The lot size shall be expressed in units of shirts. The presence of excessive defects or failure of any testing (see 6.2) shall be cause for rejection of the first article.

4.4 Conformance inspection. Unless otherwise specified in the contract (see 6.2), conformance inspection, in accordance with 3.1.2, shall consist of the examinations and tests as specified in 4.5.1 through 4.5.5.2.

4.4.1 Component and material sampling and acceptance criteria. Unless otherwise specified in referenced documents or procurement documents (see 6.2), component and material sampling shall be in accordance with Table IX. The component lot shall be unacceptable if one or more sample units fail to meet any examination or test requirements specified in this document. The unit for expressing lot size and sample unit shall be in accordance with the applicable referenced documents. For the lining, the lot size shall be expressed as yards, and the sample unit for testing shall be 3 yards, full width.

TABLE IX. Component and material sampling.

Lot size	Sample size (sample unit)
800 or less	2
801 through 22,000	3
22,001 and over	5

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4.4.2 End item sampling and acceptance criteria. Unless otherwise specified in the contract (see 6.2), sampling for inspection shall be performed for Dress/Semidress ("D/S") shirts in accordance with the provisions of MIL-STD-1492. The sample shall be one finished shirt. The lot size shall be expressed in units of shirts. The presence of excessive defects or failure of any testing as specified (see 6.2) shall be cause for rejection of the lot.

4.5 Inspection methods.

4.5.1 Component and material examinations and tests. In accordance with 4.1, components and materials shall be examined and tested in accordance with the specified examinations and test specified in Table X and Table XI unless otherwise excluded, amended, or modified in this document or applicable procurement documents (see 6.2).

TABLE X. Component and material verification.

Material	Requirement	Verification
Basic material – class 1	3.3.1.1	MIL-C-29404
Basic material – class 2	3.3.1.2	MIL-DTL-21115
Interfacing	3.3.2	A-A-59992
Interfacing, fusible - alternate	3.3.2	MIL-C-44296
Threads	3.3.4	Per specific thread specification listed in Table II
Thread, shade	3.3.4	Visual 1/
Buttons	3.3.5	A-A-59994
Buttons, shade	3.3.5	Visual 1/
Collar stays	3.3.6	Visual 2/
Label	3.3.7	MIL-DTL-32075

1/ Results recorded as pass or fail.

2/ A certificate of compliance indicating the component material meets the specified requirements may be used.

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TABLE XI. Class 2 and class 3 collar stand lining verification.

Characteristic	Requirement	Verification
Shade	3.3.3	Visual 1/
Fiber identification	3.3.3	AATCC TM20A 2/
Yarns per inch: warp and filling	3.3.3	ASTM D3775
Weight	3.3.3	ASTM D3776/D3776M
Breaking strength	3.3.3	ASTM D5034
Dry: warp and filling		
Wet: warp and filling		
Colorfastness to perspiration	3.3.3	AATCC TM15
Color change		
Staining on cotton		
Staining on wool		
Colorfastness to drycleaning	3.3.3	AATCC TM132
Color change		
Staining on cotton		
Staining on wool		
Colorfastness to crocking	3.3.3	AATCC TM8
Dimensional stability to drycleaning	3.3.3	AATCC TM158

1/ Results recorded as pass or fail.

2/ A certificate of compliance indicating the component material meets the specified requirements may be used.

4.5.2 In-process examination. Visual and dimensional examinations shall be made at any point or during any phase of the manufacturing process to determine whether construction details, which cannot be examined in the finished product, are in accordance with the requirements specified in section 3. Materials and components that contain defects and damages, as defined in the visual examination of 4.5.3, shall be removed from production. Any in-process nonconformance remaining in the finished shirt shall be classified as a defect in accordance with the visual examination of 4.5.3.

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4.5.3 Visual examination of the end item. Before the formation of a lot, each shirt shall be buttoned and examined on an inspection table for the Dress/Semidress defects identified as "D/S" specified in MIL-STD-1492. Any shirt containing a selected defect shall not be included in the end item lot. Selected defects are those defects listed in MIL-STD-1492, which are indicated by an asterisk (*) and defects specified in Table XII.

TABLE XII. Permanent crease defects.

Defect	Category
1. Permanent crease omitted, incorrectly positioned, not continuous, misaligned, crooked, wavy, angled, or not positioned vertically straight.	Major *
2. Front, pocket, and pocket flap crease lines do not create the appearance of a single, continuous line.	Major
3. Silicone bead is less than or greater than 1/16 inch.	Major
4. Evidence of flaking, decomposition, failure to retain crease, or cracking.	Major
5. Silicone not within the crease line, excessive silicone, missing silicone, voids, or not uniform application or appearance of silicone bead, staining or strike back through to the face of the material.	Major

4.5.4 Dimensional examination. The shirts shall be examined for conformance to the dimensions specified in Table VI and Table VII when measured in accordance with Table VIII (see 6.2).

4.5.5 Permanent crease testing. Shirt lots shall be tested as specified in 4.5.5.1 and 4.5.5.2 prior to the presentation of each lot. The sample unit for each specified test shall be one shirt selected at random from the end item lot. Four test specimens shall be cut for testing from each shirt: three used for testing and one used as the standard sample for comparison to determine "pass" or "fail" results of 4.5.5.2. A single result of failure in any of the specified tests shall be recorded as a lot failure.

4.5.5.1 Permanent crease resistance tests. The shirt samples shall be tested for permanent crease resistance to dry heat (3 exposures) in accordance with AATCC TM117 at a minimum of 400 °F and to steam pressing and perchloroethylene solvent (3 exposures) in accordance with AATCC TM132.

4.5.5.2 Permanency of crease test. The permanently creased test specimens shall be soaked for 30 minutes in water containing a minimum amount of wetting agent to thoroughly wet out the cloth. The temperature of the water shall be 165±5 °F. The wet out test specimens shall be removed without squeezing and shall be air dried or dried at a temperature not exceeding 180 °F while lying flat on an absorbent surface such as paper hand toweling or kitchen type tissue. The wet out and dried test specimens shall be compared with the standard sample. The sharpness of the crease on the test specimen shall be not less than that of the unwetted sample. Results shall be recorded as "pass" or "fail".

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The shirts are intended for wear by male personnel of the United States Marine Corps.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. NSNs.
c. Class and neck size and sleeve length required (see 1.2).
d. When first article is required (see 3.1.1).
e. When conformance inspection is required (see 3.1.2).
f. Applicable Government patterns (see 3.6).
g. Acceptance quality limits and inspection conditions (see 3.10, 4.3.1, 4.4.2, 4.5.3, and 4.5.4).
h. Variations to testing conditions specified in test method documents (see 4.2).
i. First article sampling and acceptance criteria (see 4.3.1).
j. Conformance inspection requirements, if other than specified (see 4.4).
k. Component and material sampling and acceptance criteria (see 4.4.1).
l. End item sampling and acceptance criteria (see 4.4.2).
m. Component and material examinations and tests, if other than specified (see 4.5.1).
n. Packaging requirements (see 5.1).

6.3 Information requests. For access to information such as purchase descriptions, patterns, drawings, and standard shade samples of cloth, go to https://www.dla.mil/TroopSupport/ClothingandTextiles/SpecRequest.aspx for the Defense Logistics Agency Troop Support's Clothing and Textile Specification/Drawing/Pattern Request Form. Complete the request form and then submit the form. Requests to use equivalent materials or components or to make changes to the pattern should be sent to the contracting officer for approval by the military services.

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6.3.1 Button color samples. Button color samples can be obtained through a request to the contracting agency.

6.4 Solution dyed acetate lining material. A solution dyed acetate lining material is recommended to meet the specified colorfastness requirements (see 3.3.2).

6.5 Thread shade. Khaki shade P-1, Cable No. 66019 of the Department of Defense (DoD) Standard Color Card for Sewing Threads (no longer updated, but still in use) or equivalent is recommended to meet the shade match requirements (see 3.3.4).

6.6 Button shade. Khaki AL, Cable No. 62026 of the DoD Standard Shades for Buttons (no longer updated, but still in use) is equivalent to shade M-2409 of the United States Marine Corps Button Shade Standard (no longer updated, but still in use) or Khaki 475 color chip 23420 of AMS-STD-595 (see 3.3.5).

6.7 NSNs. For informational purposes only, representative NSNs for items in this specification are listed in Table XIII. Table XIII is not intended to be a comprehensive list of all NSNs associated with this specification.

TABLE XIII. NSNs.

NSN	Neck size	Sleeve length
8405-01-515-9013	15	33
8405-01-515-9059	16-1/2	36

6.8 Subject term (key word) listing.

Dress
Enlisted
Male
Marine Corps
Officer
Uniform

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Navy – MC
DLA – CT

Preparing activity:
DLA – CT
(Project 8405-2024-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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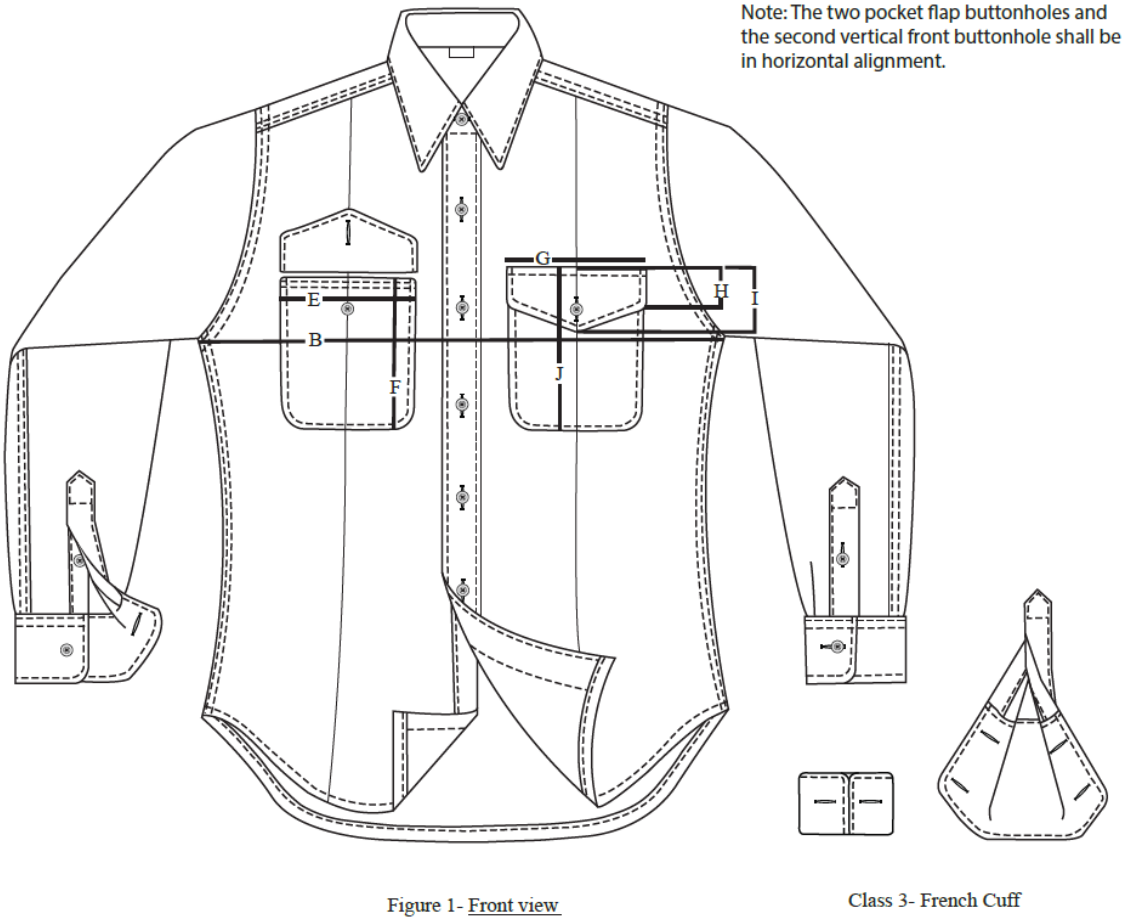


FIGURE 1. Men's long-sleeve khaki shirt – front (with measurement guidance).

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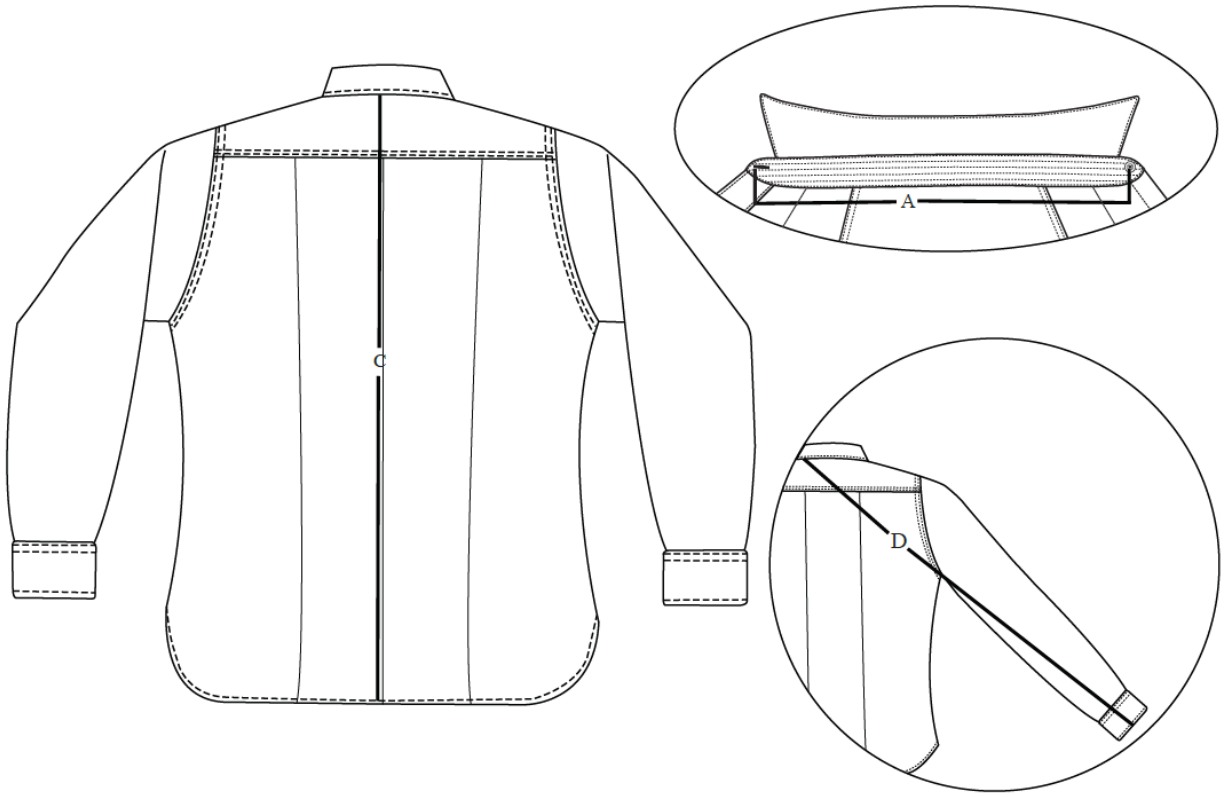


Figure 2- Back view

FIGURE 2. Men's long-sleeve khaki shirt – back (with measurement guidance).

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